Rashtreeya Sikshana Samithi Trust

R.V. College of Engineering®

(Autonomous Institution affiliated to VTU, Belagavi)

Department of Computer Science & Engineering

Bengaluru -560059

Applied Digital Logic Design & Computer Organization (CS234AI -2025 Scheme)

III Semester B.E

LABORATORY RECORD

2026 – 2027

Name of the Student:_____

U S N **:**_____

Semester **: III** **Section :**_____

R.V. College of Engineering®

(Autonomous Institution affiliated to VTU, Belagavi)

Department of Computer Science & Engineering



APPLIED DIGITAL LOGIC DESIGN & COMPUTER ORGANIZATION

(CS234AI - 2025 Scheme)

Laboratory Record (Autonomous Scheme 2025)

R.V. College of Engineering®

(Autonomous Institution affiliated to VTU, Belagavi)

Department of Computer Science & Engineering

Laboratory Certificate

This is to certify that Mr. / Ms. _____ with USN _____ of III semester has satisfactorily completed the course of experiments in Applied Logic Design and Computer Organization Lab (CS234AI) prescribed by the department during the year 2025 - 2026 .

Name of the Candidate: _____

USN No.: _____ Semester:_____

Marks	
Maximum	**Obtained**

Signature of the staff in-charge Head of the Department

Date:

Department of Computer Science and Engineering
R V College of Engineering, Bengaluru

Vision

To achieve leadership in the field of Computer Science & Engineering by strengthening fundamentals and facilitating interdisciplinary sustainable research to meet the ever-growing needs of the society.

Mission

- To evolve continually as a center of excellence in quality education in computers and allied fields.
- To develop state-of-the-art infrastructure and create environment capable for interdisciplinary research and skill enhancement.
- To collaborate with industries and institutions at national and international levels to enhance research in emerging areas.
- To develop professionals having social concern to become leaders in top-notch industries and/or become entrepreneurs with good ethics.

Program Educational Objectives (PEOs):

PEO1: Develop Graduates capable of applying the principles of mathematics, science, core engineering and Computer Science to solve real-world problems in interdisciplinary domains.

PEO2: To develop the ability among graduates to analyze and understand current pedagogical techniques, industry accepted computing practices and state-of-art technology.

PEO3: To develop graduates who will exhibit cultural awareness, teamwork with professional ethics, effective communication skills and appropriately apply knowledge of societal impacts of computing technology.

PEO4: To prepare graduates with a capability to successfully get employed in the right role and achieve higher career goals or take-up higher education in pursuit of lifelong learning.

Knowledge Profile (WK1–WK9)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (PO1–PO11)

PO1: Engineering Knowledge

Apply the knowledge of mathematics, natural science, computing, and engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis

Identify, formulate, and analyze complex engineering problems, reaching substantiated conclusions with consideration for the holistic nature of the problem.

PO3: Design/Development of Solutions

Design creative solutions for complex engineering problems and design systems, components, or processes to meet identified needs—with consideration for public health and safety, and cultural, societal, and environmental factors. Sustainability is now intrinsic to design.

PO4: Conduct Investigations of Complex Problems

Conduct investigations of complex engineering problems using research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO5: Engineering Tool Usage

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling—recognizing their limitations—to solve complex engineering problems.

PO6: The Engineer and the World

Analyze and evaluate societal and environmental aspects while solving complex engineering problems, considering their impact on sustainability with reference to economy, health, safety, legal frameworks, culture, and the environment.

PO7: Ethics

Apply ethical principles and commit to professional ethics, human values, diversity, and inclusion; and adhere to relevant national and international laws.

PO8: Individual and Collaborative Team Work

Function effectively as an individual, and as a member or leader in diverse and multi-disciplinary teams.

PO9: Communication

Communicate effectively and inclusively within the engineering community and with society at large—comprehending and writing effective reports and design documentation, and making effective presentations—considering cultural, language, and learning differences.

PO10: Project Management and Finance

Apply knowledge and understanding of engineering management principles and economic decision-making, and apply these to one's own work, as a member and leader in a team, to manage projects in multi-disciplinary environments.

PO11: Life-Long Learning

Recognize the need for, and have the preparation and ability for, independent and life-long learning, adaptability to new and emerging technologies, and critical thinking in the broadest context of technological change.

Program Specific Outcomes

PSO1: System Analysis and Design

The student will:
1. Recognize and understand the dynamic nature of developments in computer architecture, data organization and analytical methods.
2. Learn the applicability of various systems software elements for solving real-world design problems.
3. Identify the various analysis & design methodologies for facilitating development of high quality system software products with focus on performance optimization.
4. Display good team participation, communication, project management and document skills.

PSO2: Product Development

The student will:
1. Demonstrate knowledge of the ability to write programs and integrate them resulting in state-of –art hardware/software products in the domains of embedded systems, databases /data analytics, network/web systems and mobile products.
2. Participate in teams for planning and implementing solutions to cater to business – specific requirements displaying good team dynamics and professional ethics.
3. Employee state-of-art methodologies for product development and testing / validation with focus on optimization and quality related aspects

Course Outcomes (COs):

At the end of the course student will be able to

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Apply digital logic principles, Boolean algebra, arithmetic circuits, combinational and sequential logic design, and fundamental computer organization concepts to solve engineering problems.
CO2	Analyze digital systems and computer architectures by interpreting logic functions, processor operations, instruction execution, memory organization, and system performance.
CO3	Examine alternative digital design methodologies, processor organizations, memory hierarchies, and input/output mechanisms by considering functionality, efficiency, and engineering constraints.
CO4	Design reliable digital systems and processor-based solutions using appropriate digital design methodologies, hardware description languages, simulation tools, and laboratory practices
CO5	Develop digital system solutions through collaborative project-based learning by integrating digital logic design, computer organization concepts, modern engineering tools, and professional Engineering practices.

CO Mapping with WK and POs

COs	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3	3	3	1	3	-	-	-	3	3	3	1	3	-	-	-	-	-	-
CO2	2	3	3	3	1	3	-	-	-	3	3	3	2	3	-	-	-	-	-	-
CO3	2	3	3	3	1	3	-	-	-	3	3	3	2	3	1	1	1	1	1	1
CO4	2	2	3	3	1	3	-	-	-	3	3	3	2	3	1	1	1	1	1	1
C05	2	2	3	3	3	3	3	3	3	3	3	3	2	3	2	2	3	3	3	3

High-3: Medium-2 : Low-1

Course - PO Mapping

COs	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
Course	2	3	3	3	1	3	1	1	1	3	3	3	2	3	1	1	1	1	1	1

Laboratory Rules, Safety & Equipment Handling Guidelines

To ensure safe operation of laboratory equipment and effective learning, all students must strictly follow the rules given below.

---

1. Preparation Before the Lab

- Come prepared with the **digital logic circuit diagrams, IC pin details, Truthtable** for the assigned experiment.
- Read the **experiment objective, theory, and procedure** in advance.
- Bring the **lab record, data sheets, and required stationery**.
- Follow the **experiment sequence and lab schedule** as prescribed by the department.
- Ensure that you have the **required software tools or IDE familiarity** before starting the experiment (when you are using simulators / FPGA kits).

---

2. General Laboratory Rules

- Enter the laboratory **on time and with proper preparation**.
- Use **computers, digital trainer kits, FPGA kits only for academic purposes**.
- Follow **instructions given by the faculty or lab instructor** at all times.
- Work only on the **system and hardware kits assigned to you**.
- Maintain **discipline, silence, and cooperation** with other students.
- Do not move around unnecessarily during the lab session.
- Keep your **workstation clean and organized**.

3. Equipment Handling Guidelines

- Handle **digital trainer kits, FPGA kits and related cables/modules carefully**.
- Use **proper cables / patch cards / jumper wires and connectors** while making circuit connections.
- Avoid applying **excess force on pins, connectors, and ports**.
- Ensure all **components are properly connected before switching ON the power supply**.
- Return all **components and tools** to their designated storage location after use.
- Report **damaged components or malfunctioning equipment** immediately to the lab instructor.

4. Electrical & Hardware Safety

- **Do not power the circuit** until the instructor verifies the connections.
- Avoid **short circuits, reversed polarity, and incorrect power supply connections**.
- Always **switch OFF the power supply before modifying the circuit**.
- Do not touch **exposed wires or powered circuits** unnecessarily.
- Ensure proper **ground connections when interfacing sensors or modules**.
- Use **only the recommended voltage levels** for conducting the experiments

5. Computer & Network Usage Policy

- Do not **install, uninstall, or modify software** on laboratory computers.
- Do not **tamper with system files, network settings, or servers**.
- Do not **use another student's login credentials**.
- Save program files only in the **designated folder**.
- Do not **connect external storage devices (USB drives, CDs, etc.) without permission**.

6. Prohibited Activities

Students are strictly prohibited from:

- Coming **late to the laboratory or leaving early without permission**.
- **Changing the assigned system or hardware kit** without informing the lab instructor.
- **Removing any hardware component** from the laboratory.
- **Using laboratory equipment for non-academic activities**.
- **Completing the lab record instead of performing the experiment during lab time**.

7. End of the Lab Session

Before leaving the laboratory:

- Save your **program files and shut down the computer properly**.
- **Disconnect circuits carefully** and switch OFF the power supply.
- Return all **components and tools** to the designated place.
- Ensure the **workspace is clean and organized**.
- Place the **chairs back in their original position**.

Note:

Failure to follow laboratory rules and safety instructions may result in **denial of lab access or disciplinary action**.

Rubrics for ADLD & CO Laboratory

Each experiment (Part A) is evaluated for 10 marks.

Write-up and Execution rubrics (Max: 6 marks)				
		Excellent	**Good**	**Poor**
a.	**Knowledge / Understanding (2 Marks)**	Demonstrates thorough ability in describing and illustrating the function of the circuit (2-1.5)	Demonstrates considerable ability in describing and illustrating the function of the circuit (1-0.5)	Unable to demonstrate, describe and illustrate the function of the Circuit (0)
b.	**Circuit Construction (2 Marks)**	Excellent circuit functional, construction (2-1.5)	Fair circuit functional, construction (1-0.5)	Poor circuit functional, construction (0)
c.	**Circuit Data Recording (2 Marks)**	Correct data input and excellent presentation (2-1.5)	Correct data input and fair presentation (1-0.5)	Correct data input and poor presentation (0)
Viva rubrics (Max: 4 marks)				
		Excellent	**Good**	**Poor**
a.	**Circuit Data Analysis (2 Marks)**	Answers to data analysis questions, show complete understanding of how the circuit works (2-1.5)	Answers to data analysis questions, show solid understanding of how the circuit works (1)	Answers to data analysis questions, show no understanding of how the circuit works (0)
b.	**Applications (1 Mark)**	Demonstrates a high ability when wiring logic gates to construct logical circuits (1)	Demonstrates considerable ability when wiring logic gates to construct logical circuits(0.5)	Demonstrates very limited ability when wiring logic gates to construct logic circuits (0)
c.	**Communication of Ideas (1 Mark)**	Communicates all ideas clearly (1)	Communicates limited ideas (0.5)	Unable to communicate ideas(0)

PART B (EL)

Each EL to be evaluated for 20 marks.

Sl.no	**Criteria**	**Measuring Method**	**Excellent**	**Good**	**Poor**
a.	**Innovation and Design (12 Marks) CO4,CO5**	Innovativeness Design and realization	Demonstrates thorough ability in describing and illustrating the experiment (10-12M)	Demonstrates considerable ability in describing and illustrating the experiment (5-9)	Unable/Partial to demonstrate, describe and illustrate the experiment (0-4)
b.	**Viva Voce (4 Marks) CO4,CO5**	Effective Communication of the design, Understanding of basics	Excellent understanding of the design (4)	Fair understanding of the design (2-3)	Poor understanding of the design (0-1)
c.	**Report and presentation (4 Marks) CO4,CO5**	Documentation and Presentation skills	Correct data input and excellent presentation (4)	Correct data input and fair presentation (2-3)	Correct data input and poor presentation (0-1)

PART-A : List of Laboratory Experiments
[Conducted using Digital Trainer kit, Simulator and FPFA kit] [Record-20M,Test-10M]

Sl. No	Contents	Date	Page No.	Marks (10)			Staff Sign
				Record (06)	Viva (04)	Total (10)	
	PART A						
1	Realization of BCD to Excess-3 Code converter with Parallel Adder and Subtractor using 4 bit Adder, using the IC – 7483.		14				
2	Realization of Full Adder and Full Subtractor using Multiplexers, using the IC 74153.		17				
3	Design and realization One Bit and Two-Bit Magnitude Comparator using Basic Gates.		22				
4	a.Realization of Binary to Gray Code Converter and vice-versa using IC 74139. b.Realization of single digit Seven segment display using the BCD to seven segment decoders, using the IC–7447 and Realization of Priority Encoder using IC -74147		26				
5	Design and Realization of Master-Slave JK Flip Flop using NAND Gate		33				
6	a.Realization of Up and Down Programmable Counter using IC74192. b.Design a 4bit Universal Shift Register, Verify SISO, SIPO, PISO, PIPO operations. Realize the Ring Counter and Jhonson Counter using the SR. Use the Simulator		36				
7	Design of Mod-N Synchronous counters using JK/D/T Flip-Flops Using Simulator		43				
8	4bit CPU Design: ALU and Register File Design		47				
9	4bit CPU Design: Memory Interface, Decoding & Control Unit		47				
10	Instruction Set Preparation and ARM Assembly Programs.		47				
11	Realize the 4bit CLA & 4bit Comparator using HDL/Verilog on FPGA		48				
12	Evaluate & compare the different Memory & I/O standards on PC Motherboard		48				
					Total Marks	**120**	
Reduced Part A Marks = (Total Marks*20)/ 120 = /20							

Laboratory Test: Writeup= /2M,Conduction= /6M,Viva= /2M:- /10M

PART-B [Experiential Learning / Project Based Learning] [20Marks]

EL							
Develop any one of the following : 1. Microcontrollers based Embedded System 2. Assembly Language based software application 3. Digital System Design using VHDL							
Name of the EL Project	Date	Page No	Maximum Marks (10)				Staff Sign
			Design (12)	Viva (4)	Report (4)	Total (20)	

LAB INTERNALS		
RECORD (Part-A)	**Max – 20**	
TEST (Part-A)	**Max - 10**	
PART B (EL)	**Max - 20**	
TOTAL	**Max – 50**	
Signature of the faculty		

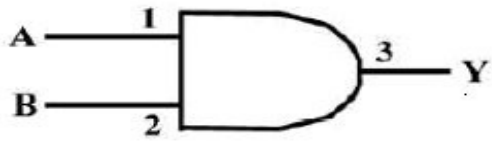
Introduction to Logic gates

This chapter is intended to introduce the student to some of the basic gates that will be used in the foundations of computer systems design laboratory experiments. The content of this chapter includes the Pin diagram of basic gates. Following are the basic gates that are dealt in this chapter.

1. IC 7408: 2- input AND gate
2. IC 7400: 2-input NAND gate
3. IC 7486: X-OR gate
4. IC 7432: 2-input OR gate
5. IC 7404: NOT gate
6. IC 7402: 2-input NOR gate
7. IC 7410: 3-input NAND gate
8. IC 7411: 3-input AND gate
9. IC 7420: dual 4-input NAND gate
10. IC 7421: dual 4-input AND gate

All the above-mentioned IC`s are known as small-scale integrated circuits which belongs to a logical family known as TTL.

IC7408: Two Input AND Gate



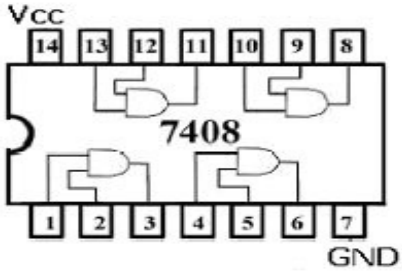
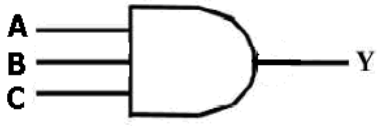


Fig.1.1 (a) AND Gate Logic Symbol **(b) Pin Diagram for the 7408 Quad AND**

IC7411: Three Input AND Gate



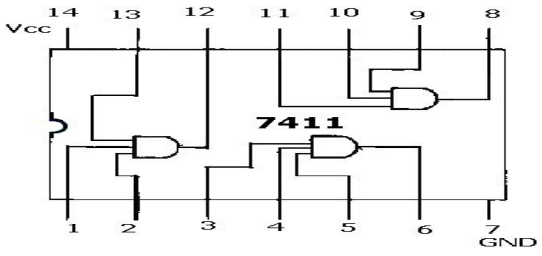


Fig.1.2 (a) Logic Symbol for three input AND gate **(b) Pin Diagram for the 7411 Triple 3 input AND gate**

IC7432: Two Input OR Gate



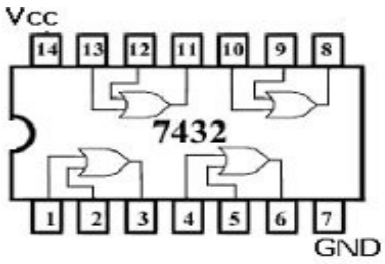
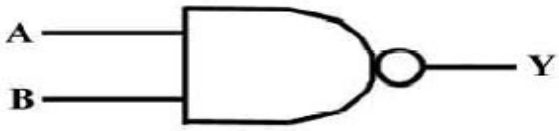


Fig.1.3 (a) OR Gate Logic Symbol **(b) Pin Diagram for the 7432 Quad OR**

IC 7400: Two Input NAND Gate



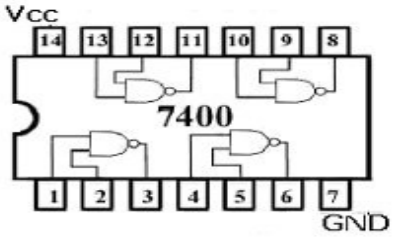


Fig.1.4 (a) NAND Gate Logic Symbol **(b) Pin Diagram for the 7400 Quad NAND**

IC 7410: Three Input NAND Gate



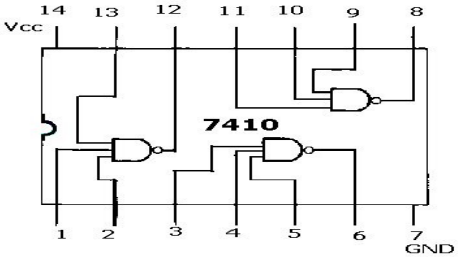
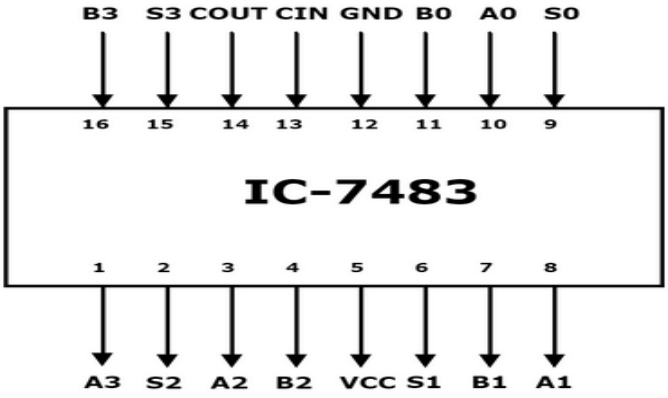


Fig.1.5 (a) Logic Symbol Of three input NAND **(b) Pin Diagram for the 7410 Triple 3 input NAND gate**

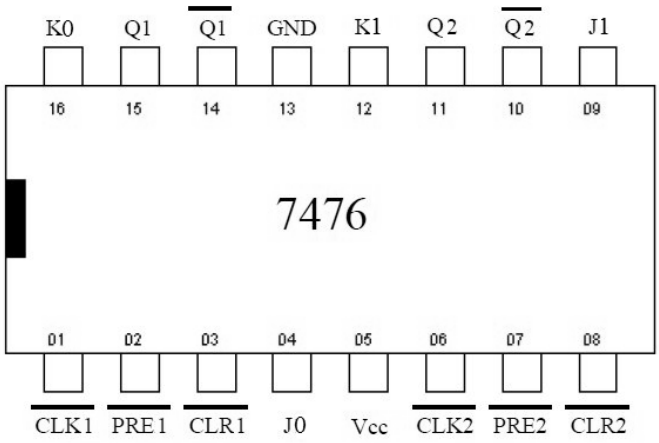
IC 74283 Parallel Adder



IC 7476: J K Flip-flop

IC 7476 : The 74LS76 is a Dual JK flip-flop IC, which contains two individual JK type bi-stable within a single chip enabling single or master-slave toggle flip-flops to be made. This is a negative-edge triggered flip-flop, with both active low preset and clear inputs.

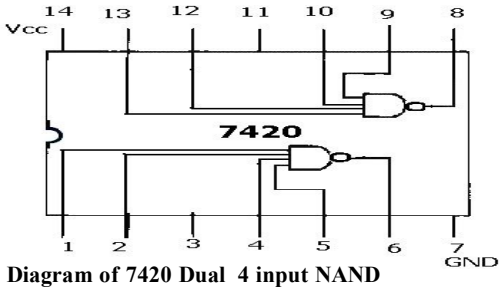
.



IC 7420: Four Input NAND Gate



Fig.1.6 (a) Logic Symbol Of four input NAND gate gate



(b) Pin Diagram of 7420 Dual 4 input NAND

IC 7402: Two Input NOR Gate

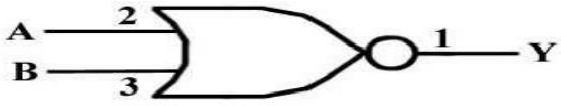
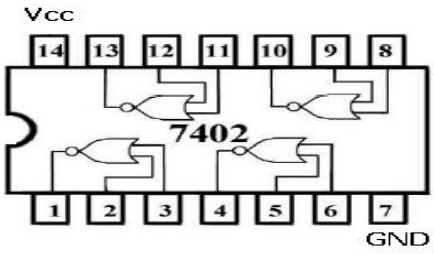


Fig.1.7 (a) NOR Gate Logic Symbol NOR



(b) Pin Diagram for the 7402 Quad

IC7486: Two Input EX-OR Gate

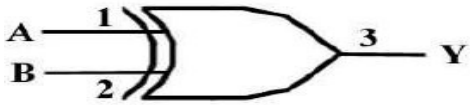
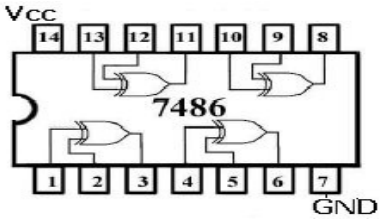


Fig.1.8 (a) X-OR Gate Logic Symbol



(b) Pin Diagram for the 7486 Quad X-OR

IC7404: Two Input NOT Gate

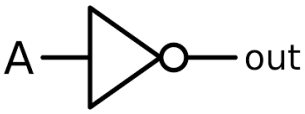
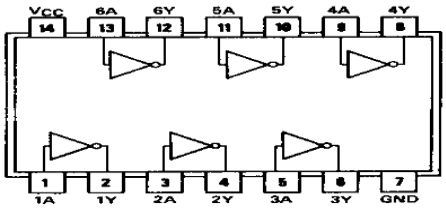


Fig: 1.9 (a) Pin Diagram for the 7404 NOT Gate Inverter



(b) Pin Diagram for the 7404

Experiment 01

1. Realization of Excess-3 Code converter with Parallel Adder and Subtractor using IC – 7483.

Aim: To design Excess-3 Code converter using IC 74283 parallel adder and verify the truth table

Components Required:

Sl. No.	Components	Specification	Quantity
1	Parallel Adder	IC -7483	1 No
2	2 Input EX-OR	IC -7486	1 No
3	Digital trainer kit & patch cables	–	1 No, 20 patch cables

Procedure: -

1. Verify the gates.
2. Make the connections as per the circuit diagram.
3. Connect to VCC and apply various combinations of input according to the truth table.
4. Note down the output readings for different combinations of inputs.

Theory:

The availability of large variety of codes for the same discrete elements of information results in the use of different codes by different systems. A conversion circuit must be inserted between the two systems if each uses different codes for same information. Thus, code converter is a circuit that makes the two systems compatible even though each uses different binary code.

Parallel adders are digital circuits that compute the addition of variable binary strings of equivalent or different size in parallel. The IC 7483 is 4 bit binary adder, adds two 4bit binary numbers. numbers being processed are A4 A3 A2 A1and B4 B3 B2 B1, while the answer is S4 S3 S2 S1. Adder can be used as subtractor, by feeding 2s compliment of 2nd number (B). This is done by producing 1s compliment (using inverters- exor gates are used to selectively invert the input) and then adding logic 1 (by feeding logic 1 at the carry input). The circuit shown below, acts as adder / subtractor, based on the control input fed to one of the inputs of EX-OR gate. If control input is logic 0, circuit acts as adder, if control input is logic 1, circuit acts as subtractor.

Note: When used as subtractor, always remember, the carry produced is opposite of what is expected. (compliment of the actual carry).

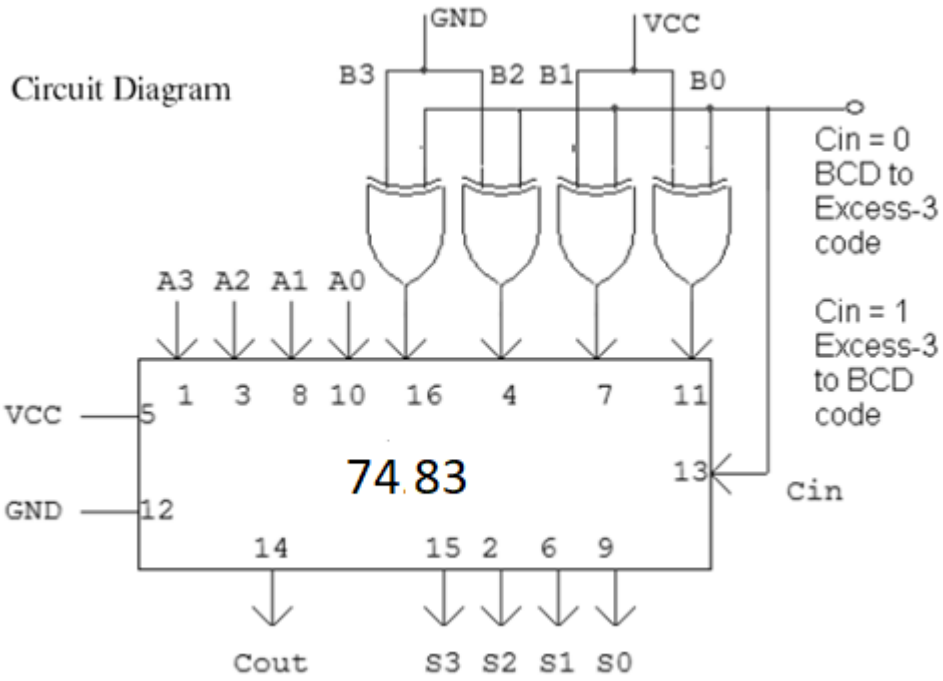
Binary Coded Decimal:

Binary Coded Decimal is a method of using binary digits to represent the decimal digits 0 through 9. It is possible to assign weights to the binary bits according to their positions. The weights in the BCD code are 8, 4, 2 and 1. Ex: $(137)_{10}$ - BCD equivalent $(0001\ 0011\ 0111)_{2.}$

Excess-3 Code:
This is an un-weighted code. Its code assignment is obtained from the corresponding value of BCD after the addition of $(0011)_2$.

BCD to Excess-3 (or) Excess-3 to BCD: Since each code uses four bits to represent a decimal digit, there must be four inputs and four output variables. The input variable are designated as B3, B2, B1, B0 and the output variables are designated as E3 , E2, E1, E0 in the truth table.

Circuit Diagram:



A. Perform addition and subtraction for following set of numbers (observe o/ps)

Example1(add): 40 and 59

Example2(add): 79h and AFh (in hexadecimal) (0x79 and 0xAF)

Example3(sub): 67 – 48

Example4(sub): 0xA3 – 0xBC

B. Code Converter (BCD-Excess 3)

Truth Table:

i) BCD - EXCESS-3 CODE

BCD				EXCESS-3			
0	0	0	0				
0	0	0	1				
0	0	1	0				
0	0	1	1				
0	1	0	0				
0	1	0	1				
0	1	1	0				
0	1	1	1				
1	0	0	0				
1	0	0	1				

ii) Excess-3 – BCD CODE

EXCESS-3				BCD			
0	0	1	1				
0	1	0	0				
0	1	0	1				
0	1	1	0				
0	1	1	1				
1	0	0	0				
1	0	0	1				
1	0	1	0				
1	0	1	1				
1	1	0	0				

Inference:

Experiment 02

Realization of Full Adder and Full Subtractor using IC 74153.

A. **Aim:** To Realize combinational circuit Full Adder using IC74153

Components Required:

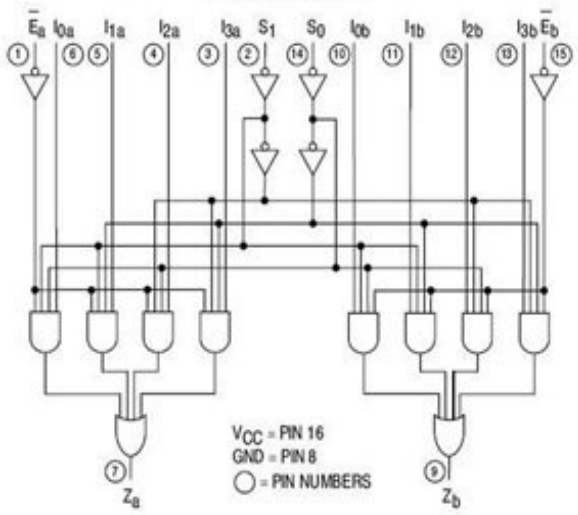
Sl. No.	Components	Specification	Quantity
1	Multiplexer	IC -74153	1 No
2	NOT GATE	IC -7404	1No.
3	Digital trainer kit & patch cables	–	1 No, 20 patch cables

Theory

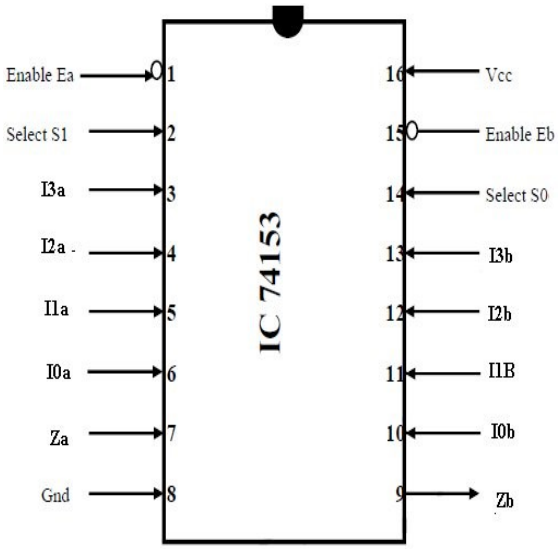
IC-74153 is a dual 4-to-1 multiplexer have 2 sets of 4:1 MUX name MUX A and MUX B, and the IC has 2 select lines S0 and S1 for each of the MUX but these selection line are internally shorted together as shown in the logic diagram below and hence the entire IC has only two selection pins. Each 4:1 MUX is associated with an active low enable pin namely E_a and E_b. With appropriated logic level on enable pins and on selection pins, logic levels on the input pin of any one or both the MUX can be transferred to the output. Z_a and Z_b are the output of MUX A and MUX B respectively.

74LS153 Dual 4-Input Multiplexer

Logical Diagram



Pin Diagram IC 74153



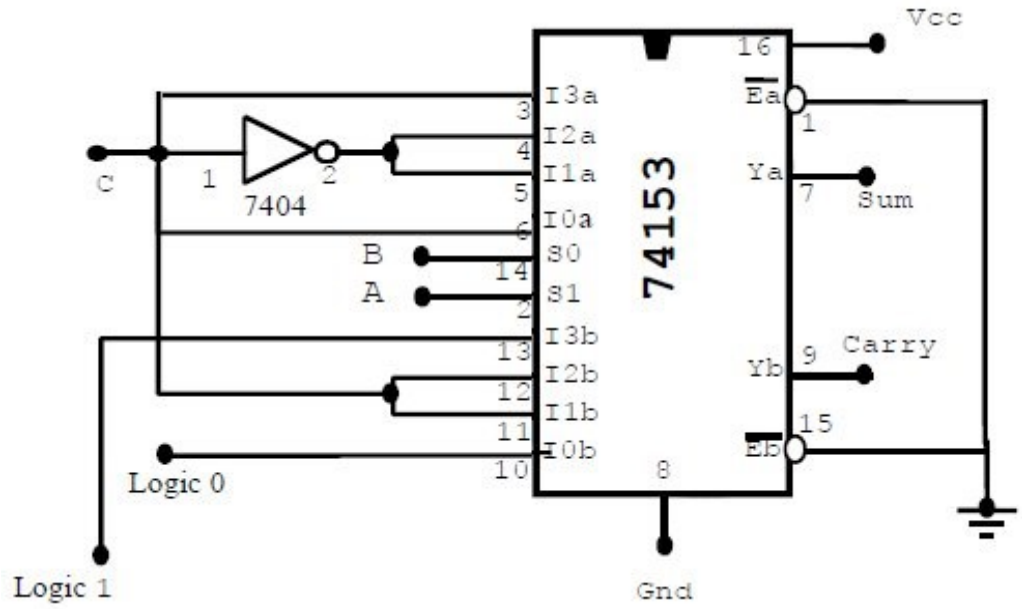
To realize any Boolean function with ‘ n’ input variable, MUX require n-1 or n-2 selection pins. We use either Tabular method or Floyd method to generate the expressions. Of these two methods tabular method is been used in this experiment to realize Adder and Subtractor using 4:1 MUX.

In this experiment**,** Sum is realized on Mux A, Carry realized on Mux B, both enable pins of the Mux are connecting to logic 0, Connect A,B inputs to S1, S0 lines of Mux respectively. According to input A and B different data inputs lines (I0 to I3) are selected. Connect the data inputs to logic 0 or logic 1 according to the required outputs. Expression for sum and carry in terms input variable C, as shown below diagrams.

Truth table of Full adder:

Inputs			**Outputs**	
A	**B**	**C**	Sum	Carry
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Logic Diagram:



For Sum:

C	A=0, B=0	A=0, B=1	A=1, B=0	A=1, B=1	Remark
0	0	1	1	0	Required Output
1	1	0	0	1	
	I0a=C	I1a=C'	I2a=C'	I3a=C	Input to selected lines

For Carry:

C	A=0, B=0	A=0, B=1	A=1, B=0	A=1, B=1	Remark
0	0	0	0	1	Required Output
1	0	1	1	1	
	I0b=0	I1b=C	I2b=C	I3b=1	Input to selected lines

Procedure:

1. Verified IC Number before conducting experiment
2. The IC is fixed into IC socket of the trainer kit and connects VCC and GND.
3. Connections are made as shown in logic diagram.
4. Connect all the inputs to switches and out to the LED's
5. The inputs are applied using the toggle switches from the trainer kit according to the truth table and the corresponding outputs are verified on the LEDs.
6. Truth table is verified for different combination of input.

Inference:

B. Aim: To Realize a Full Subtractor using IC74153

Components Required:

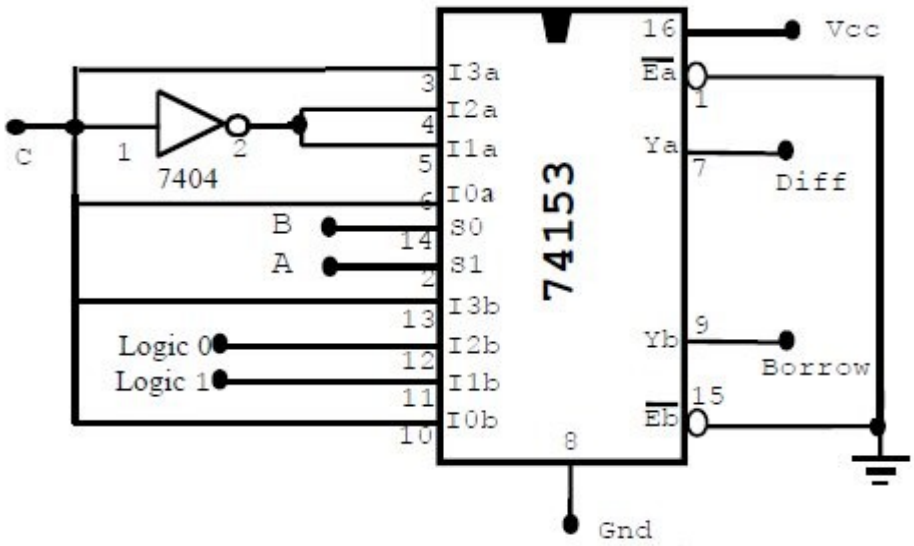
Sl. No.	Components	Specification	Quantity
1	Multiplexer	IC -74153	1 No
2	NOT GATE	IC -7404	1No.
3	Digital trainer kit & patch cables	–	1 No, 20 patch cables

Full Subtractor using 74153

Truth table of Full subtractor:

Inputs			Outputs	
A	**B**	**C**	Diff	Borrow
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Logic Diagram:



For Difference

C	A=0, B=0	A=0, B=1	A=1, B=0	A=1, B=1	Remark
0	0	1	1	0	Required Output
1	1	1	0	1	
	I0b=C	I1a=C'	I2a=C'	I3a=C	Input to selected lines

For Borrow:

C	A=0, B=0	A=0, B=1	A=1, B=0	A=1, B=1	Remark
0	0	1	0	0	Required Output
1	1	1	0	1	
	I0b=C	I1b=1	I2b=0	I3b=C	Input to selected lines

Procedure:

1. Verified IC Number before conducting experiment
2. The IC is fixed into IC socket of the trainer kit and connects VCC and GND.
3. Connections are made as shown in logic diagram.
4. Connect all the inputs to switches and out to the LED's
5. The inputs are applied using the toggle switches from the trainer kit according to the truth table and the corresponding outputs are verified on the LEDs.
6. Truth table is verified for different combination of input.

Inference:

Assigment 3: Realize 8:1 Mux using 2:1 & 4:1 Mux, 16:1 using 8:1 & 4:1

Experiment 03

Design and Realization of One and Two Bit Magnitude Comparator using Basic Gates

A. Aim: To design one bit comparator, two bit comparator using logic gates

Components Required:

Sl. No.	Components	Specification	Quantity
1	AND GATE	IC -7408	1 No
2	NAND GATE	IC-7400	1 No.
3	3 Input NAND GATE	IC-7410	1 No.
4	NOT GATE	IC -7404	1 No.
5	XOR GATE	IC-7486	1 No.
6	Digital trainer kit & patch cables	–	1 No, 20 patch cables

Theory

One Bit Comparator

A magnitude comparator is a combinational circuit that compares two numbers A and B and determines their relative magnitude. The outcome of the comparison is specified by 3 binary variables that indicate whether **A = B** or A**< B** or **A>B**.

Truth table for one bit Comparator

Inputs		Outputs		
A	**B**	**A<B**	**A=B**	**A>B**
0	0			
0	1			
1	0			
1	1			

Boolean Expression (A<B) =

Boolean Expression (A=B) =

Boolean Expression (A>B) =

K Map (A<B)

A \ B	0	
0		
1		

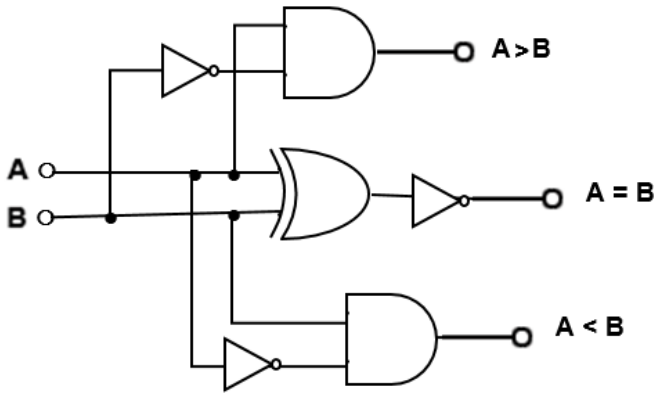
K Map (A=B)

	0	1
0		
1		

K Map (A>B)

A \ B	0	1
0		
1		

Logic Diagram of One Bit Comparator:



Two Bit Comparator

A magnitude comparator is a combinational circuit that compares four digital Bits A1, A0 and B1, B0 and determines their relative magnitude. The outcome of the comparison is specified by 3 binary variables that indicate whether A = B or A < B or A >B.

Truth table for Two bit Comparator

Observations:

Inputs				**Outputs**		
A1	**A0**	**B1**	**B0**	A<B	A=B	A>B
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

K-Map for (A<B)

Boolean Expression

(A<B)

A1 A0 \ B1 B0	00	01	11	10
00				
01				
11				
10				

Boolean Expression (A=B)

K-Map for (A=B)

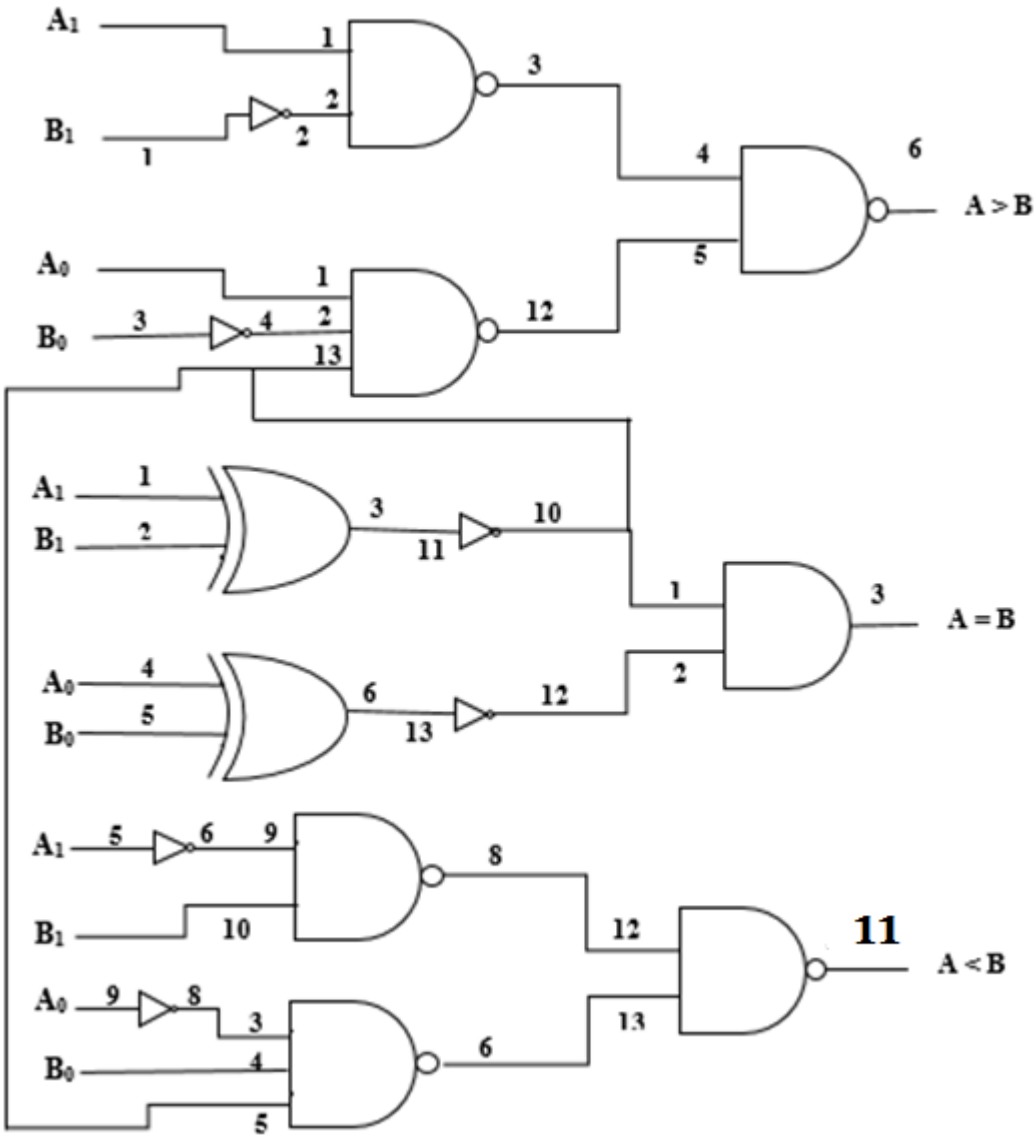
A1 A0 \ B1 B0	00	01	11	10
00				
01				
11				
10				

Boolean Expression (A>B)

K-Map for (A>B)

A1 A0 \ B1 B0	00	01	11	10
00				
01				
11				
10				

Logic diagram of two-bit comparator:



Procedure:

1. Verified IC Number before conducting experiment
2. The IC is fixed into IC socket of the trainer kit and connect VCC and GND.
3. Connections are made as shown in logic diagram.
4. Connect all the inputs to switches and out to the LED's
5. The inputs are applied using the toggle switches from the trainer kit according to the truth table and the corresponding outputs are verified on the LEDs.
6. Truth table is verified for different combination of input.

Inference:

Experiment 04

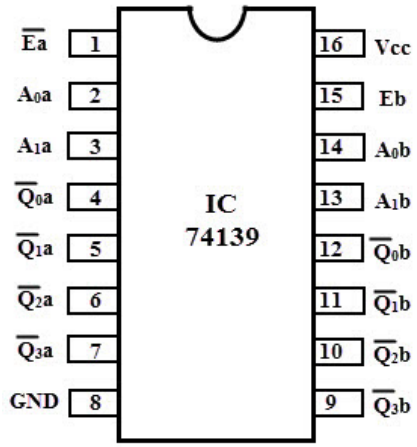
A. Realization of Binary to Gray Code Converter and vice-versa using IC 74139
B. Realization of Decoder using IC-7447 and Realization of Encoder using IC-74147

A. Aim : Realization of Binary to Gray Code Converter and vice-versa using IC 74139
Components Required:

Sl. No.	Components	Specification	Quantity
1	2-to-4 Line Decoder	74139	1 No
2	4 Input NAND gate	7420	1 No
3	Digital trainer kit & patch cables	–	1 No, 20 patch cables

Theory

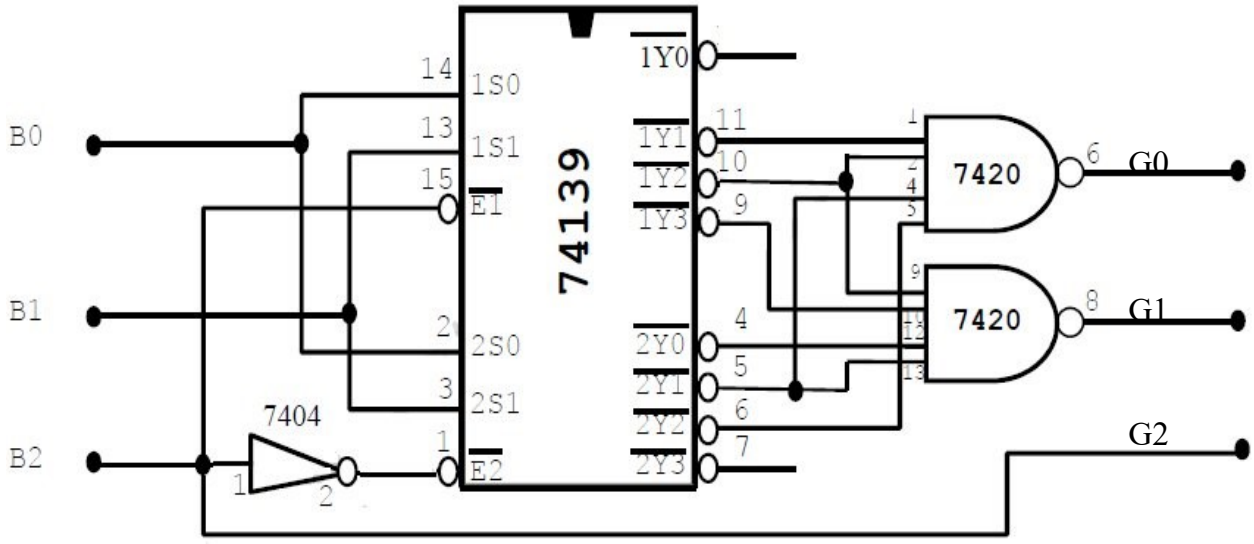
IC 74139: This IC is a dual 2-to-4-line decoder in the 7400 series. Each side of the chip is a decoder with an active low enable from a 2-bit address to four active low signals. The two decoders can be combined to realize a 3-8 decoder. This IC can also function as a dual 2 x 4 DEMUX also. If it is used as DEMUX, then active low enable pin acts as i/p of mux, 2 inputs acts as select lines of mux and 4 output lines in each decoder becomes outputs.



Procedure: -

1. Verify IC Number before conducting experiment
2. The IC is fixed into IC socket of the trainer kit and connect pin 16 to VCC and pin 8 to GND.
3. Connections are made as shown in logic diagram.
4. Connect all the inputs to switches and out to the LED's
5. The inputs are applied using the toggle switches from the trainer kit according to the truth table and the corresponding outputs are verified on the LEDs.
6. Truth table is verified for different combination of input.

Circuit Diagram:



Truth Table: 3 Bit binary code conversion using IC74139

Inputs			Outputs		
B2	**B1**	**B0**	**G2**	**G1**	**G0**
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

G0 = ∑ m( )

G1 = ∑ m( )

G2 = ∑ m( ) = B2

Note: Students are required to realize given SOP & POS expressions using the decoder IC 74139 (document them in the data sheet).

Inference:

B. Realization of Decoder using IC-7447 and Realization of Encoder using IC-74147

1. Aim: To realize a Decoder using IC-7447 and Encoder using IC-74147

Components Required:

Sl. No.	Components	Specification	Quantity
1	Decoder	IC -7447	1 No
2	7 Segment display LED		1 No.
3	Digital trainer kit & patch cables	–	1 No, 20 patch cables

Theory:

74LS47 is a BCD to 7-segment decoder/driver IC. It accepts a binary coded decimal as input and converts it into a pattern to drive a seven-segment for displaying digits 0 to 9. Binary coded decimal (BCD) is an encoding in which each digit of a number is represented by its own binary sequence (usually of four bits).
For example 239 in BCD is represented as 0010 0011 1001.

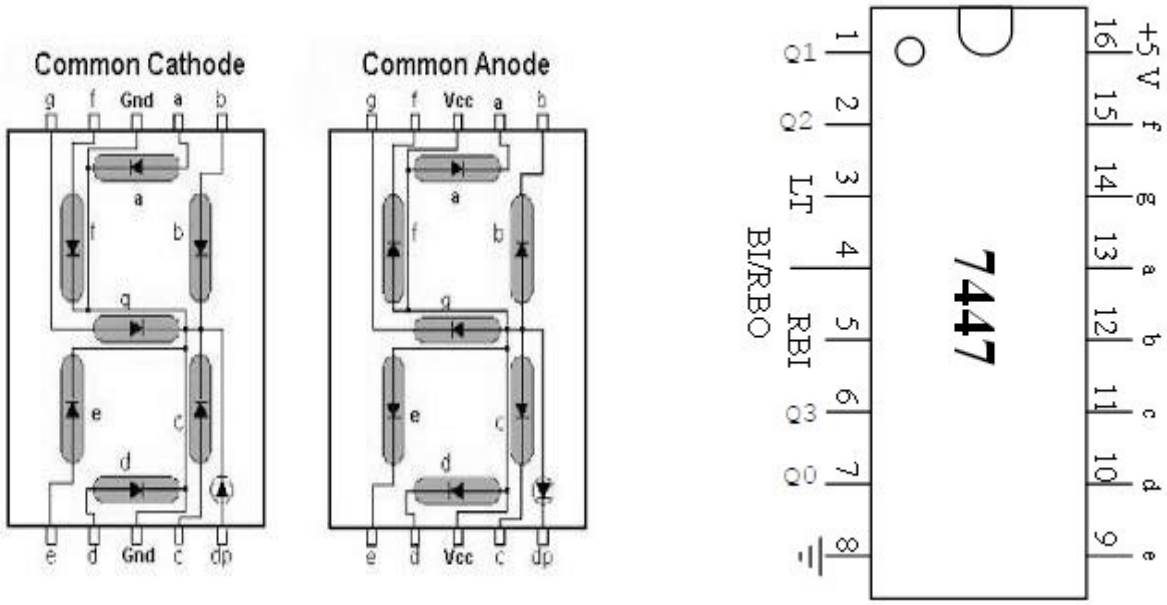
74LS47 IC accepts four lines of BCD (8421) input data and generates their complements internally. The data is decoded with seven AND/OR gates to drive indicator LEDs of the seven segment directly. The outputs correspond to Common anode (CA) configuration of seven segment. The 74LS47 chip is used to drive 7 segment display. You must use the 74LS47 with a **common anode 7-segment display** (e.g. King bright part number SA03). The input to the 74LS47 is a binary number **DCBA** where D is 8s, C is 4s, B is 2s and A is 1s. The inputs DCBA often come from a binary counter. The display is only sensible if the binary number is between DCBA=0000 (0) and DCBA=1001 (9); this is called Binary Coded Decimal or BCD for short.

The inputs / $\overline{BI}$ $\overline{RBO}$, $\overline{RBI}$ $\overline{LT}$ and (LAMP TEST)are usually connected to +5v.

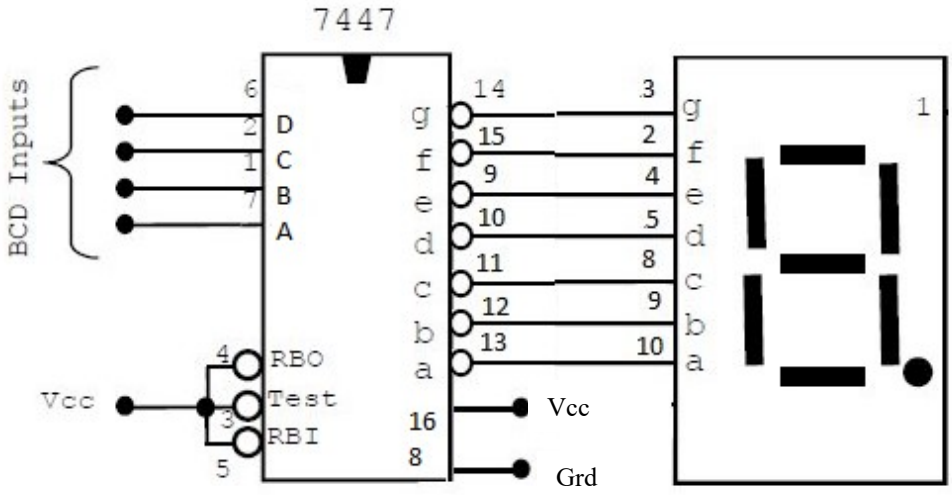
Procedure

1. Connect V_{CC} [pin 16], $\overline{LT}$ $\overline{BI}$ $\overline{RBO}$ $\overline{RBI}$ [pin 3], / [pin 4] and [pin 5] to 5v.
2. Connect GND [pin 8] to 0v.
3. Connect DCBA [pins 1, 2, 6 and 7] to DCBA on your counter.
4. Connect [pins 9-15] to abcdefg on the **common anode** 7-segment display.
5. Apply BCD input on DCBA lines using toggle switches and observe the output on7-segment LED display unit

Pin Diagram IC 7447



Using 7447 (Decoder) and seven segment decoder Logic diagram



Truth Table:

Binary Inputs	Decoder Outputs	7 Segment Display Outputs
D C B A	a b c d e f g	
0 0 0 0	0 0 0 0 0 0 1	0
0 0 0 1	1 0 0 1 1 1 1	1
0 0 1 0	0 0 1 0 0 1 0	2
0 0 1 1	0 0 0 0 1 1 0	3
0 1 0 0	1 0 0 1 1 0 0	4
0 1 0 1	0 1 0 0 1 0 0	5
0 1 1 0	0 1 0 0 0 0 0	6
0 1 1 1	0 0 0 1 1 1 1	7
1 0 0 0	0 0 0 0 0 0 0	8
1 0 0 1	0 0 0 0 1 0 0	9

Procedure:

1. Verified IC Number before conducting experiment
2. The IC is fixed into IC socket of the trainer kit and connects VCC and GND.
3. Connections are made as shown in logic diagram.
4. Connect all the inputs to switches and out to the LED's
5. The inputs are applied using the toggle switches from the trainer kit according to the truth table and the corresponding outputs are verified on the LEDs.
6. Truth table is verified for different combination of input.

Inference:

B. Aim: To realize an Encoder using IC 74147

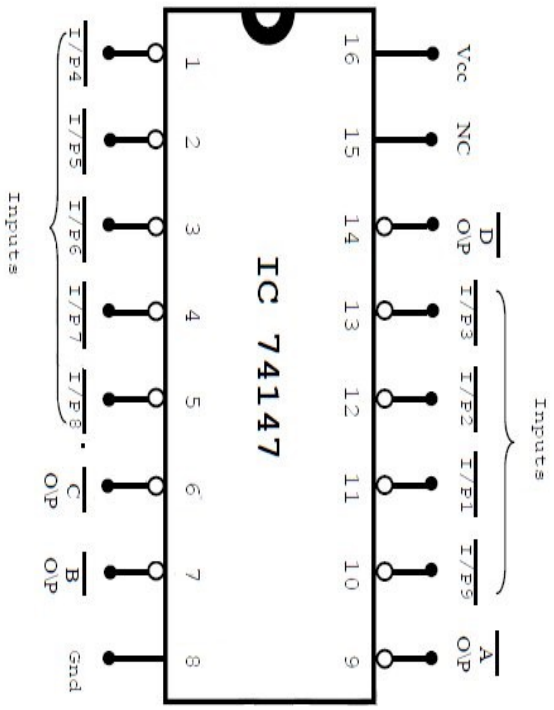
Components Required:

Sl. No.	Components	Specification	Quantity
1	Encoder	IC -74147	1 No
2	Digital trainer kit & patch cables	–	1 No, 20 patch cables

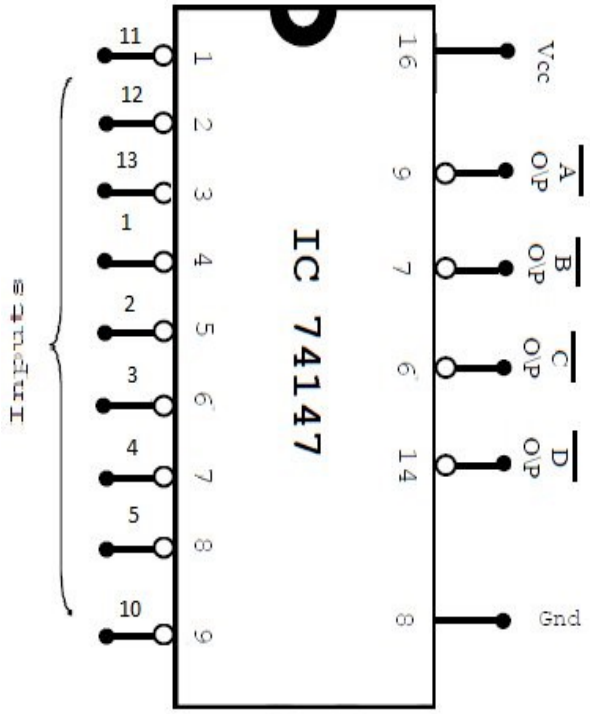
Theory:
IC 74147

The 74HC/HCT147 is 9-input priority encoders, accept data from nine active LOW inputs A 0 to A8 and provide a binary representation on the four active LOW outputs Y 0 - Y3. A priority is assigned to each input so that when two or more inputs are simultaneously active, the input with the highest priority is represented on the output, with input line A8 having the highest priority. The device provide the 9 to 4-line priority encoding function by use of the implied decimal "zero". The "zero" is encoded when all nine data inputs are HIGH, forcing all four outputs HIGH.

Pin Diagram



Logic Diagram



Truth Table:

Active – Low Decimal Inputs									Active Low BCD Output			
1	2	3	4	5	6	7	8	9	D	C	B	A
1	1	1	1	1	1	1	1	1				
0	1	1	1	1	1	1	1	1				
X	0	1	1	1	1	1	1	1				
X	X	0	1	1	1	1	1	1				
X	X	X	0	1	1	1	1	1				
X	X	X	X	0	1	1	1	1				
X	X	X	X	X	0	1	1	1				
X	X	X	X	X	X	0	1	1				
X	X	X	X	X	X	X	0	1				
X	X	X	X	X	X	X	X	0				

Procedure:

1. Verify IC Number before conducting experiment
2. The IC is fixed into IC socket of the trainer kit and connect VCC and GND.
3. Connections are made as shown in logic diagram.
4. Connect all the inputs to switches and out to the LED's
5. The inputs are applied using the toggle switches from the trainer kit according to the truth table and the corresponding outputs are verified on the LEDs.
6. Truth table is verified for different combination of input.

Inference:

Experiment 05

Realization of Master-Slave J K Flip-Flop Using NAND gates.

Aim: To realize Master-Slave JK Flip-Flops using NAND Gates and IC 7476

Components Required:

Sl. No.	Components	Specification	Quantity
1	Flip-Lop	IC -7476	1 No
2	NAND GATE	IC-7400	1 No
3	3 INPUT NAND GATE	IC-7410	2 No
4	Digital trainer kit & patch cables	_	1 No, 35 patch cables

Theory:

Flip-Flop

A flip-flop circuit can maintain a binary state indefinitely (as long as power is delivered to the circuit) until directed by an input signal to switch states. A flip-flop has two useful states, when Q=1 and Q' = 0, it is in the set state (or 1-state). When Q = 0 and Q' = 1, it is in the clear state (or 0-state). The outputs Q and Q' are complements of each other and are referred to as the normal and complemented outputs, respectively. The binary state of the flip-flop is taken to be the value of the normal output.

The NAND basic flip-flop circuit operates with both inputs normally at 1 unless the state of the flip-flop has to be changed. The application of a momentary 0 to the set input causes output Q to go to 1 and Q' to go to 0, thus putting the flip-flop into the set state. After the set input returns to 1, a momentary 0 to the reset input causes a transition to the clear state. When both inputs go to 0, both outputs go to 1 – a condition avoided in normal flip-flop operation.

JK Flip-Flop

A JK flip-flop is a refinement of the RS flip-lop in that the indeterminate state of the RS type is defined in the JK type. Inputs J and K behave like inputs S and R to set and clear the flip-flop (note that in a JK flip-flop, the letter J is for set and the letter K is for clear). When inputs are applied to both J and K simultaneously, the flip-flop switches to its complement state, that is, if Q=1, it switches to Q=0, and vice versa.

Master – Slave Flip-flop:

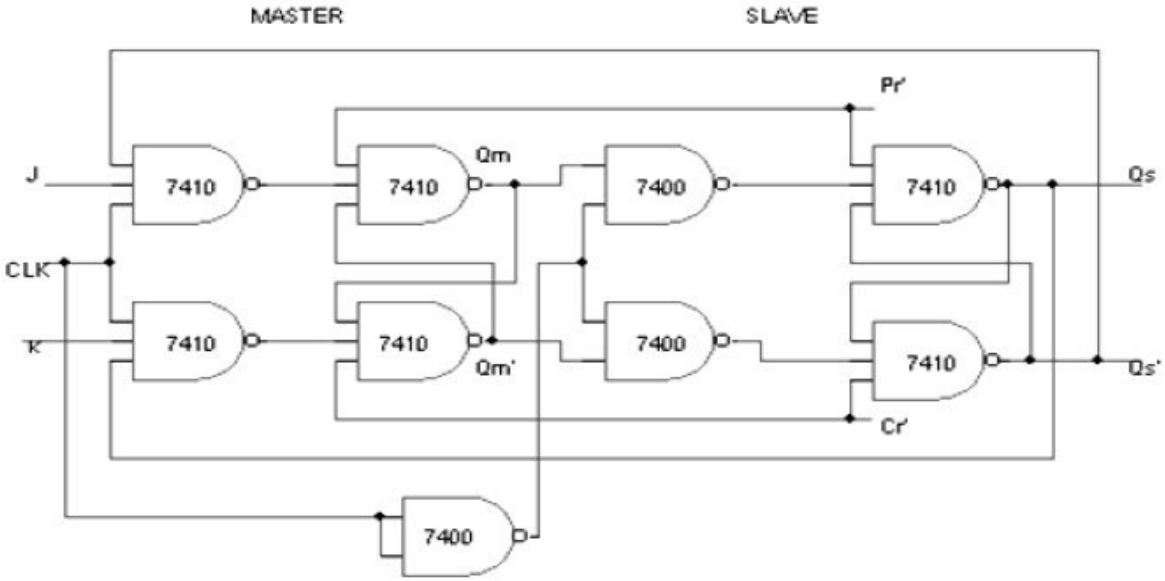
A master – slave flip-flop is constructed from two separate flip-flops. One circuit serves as a master and the other as a slave, and the overall circuit is referred to as a master-slave flip -flop. It consists of a master flip-flop and a slave flip-flop. When clock pulse CP is 0, the output of the inverter is 1. Since the clock input of the slave is 1, the flip-flop is enabled and output Q is equal to Y, while Q' is equal to Y'. The master flip flop is disabled because CP = 0. When the pulse becomes 1, the information then at the external R and S inputs is transmitted to the master flip-flop. The slave flip-flop, however, is isolated as long as the pulse is at its 1 level, because the output of the inverter is 0. When the pulse returns to 0, the

master flip-flop is isolated, which prevents the external inputs from affecting it. The slave flip-flop then goes to the same state as the master flip-flop.

Procedure:

1. Rig up the circuit as shown in the figure
2. Keep both preset and reset/clear at logic high
3. Verify the truth table for JK Master-Slave

Logic Diagram of JK Master Slave using NAND Gate



Truth Table:

CLK	J	K	Qn+1(next state)
H	0	0	
H	0	1	
H	1	0	
H	1	1	

Note: Observe the output behavior with Preset and Clear inputs and prepare the modified truth table.

Assignment: Realize D and T flip-flops using above designed MS-JK flip-flops.

Inference

Experiment 06

a. Realization of Up and Down Programmable Counter using IC74192 Aim: To Realization of counter using IC74192

(i) Up / Down counter
(ii) Presettable Up counter
(iii) Presettable Down counter

Components Required:

Sl. No.	Components	Specification	Quantity
1	COUNTER	IC -74192	1 No
2	NOT GATE	IC-7404	1 No
3	4 INPUT NAND GATE	IC-7420	1 No
4	Digital trainer kit & patch cables	–	1 No, 20 patch cables

IC74192:

74HC192: 74HCT192 is a 4-bit synchronous decade binary up/down counter. Separate up/down clocks, ClkU and ClkD respectively, simplify operation. The outputs change state synchronously with the LOW-to-HIGH transition of either clock input. If the CLKU clock is pulsed while CLKD is held HIGH, the device will count up. If the CLKD clock is pulsed while CLKU is held HIGH, the device will count down.

Only one clock input can be held HIGH at any time to guarantee predictable behavior. The device can be cleared at any time by the asynchronous master reset input (Clr); it may also be loaded in parallel by activating the asynchronous parallel load input.

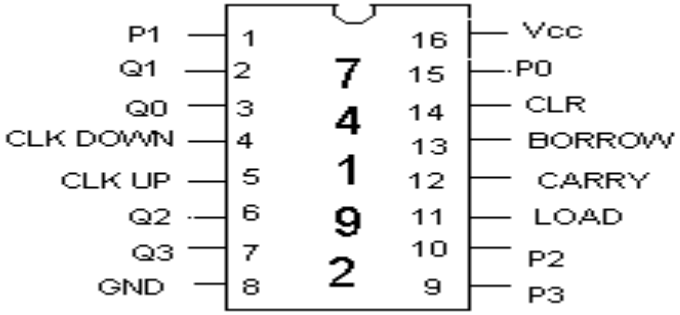
The terminal count up (Carry) and terminal countdown (Borrow) outputs are normally HIGH. When the circuit has reached the maximum count state of 09, the next HIGH-to-LOW transition of CLKU will cause carry to go LOW. Carry will stay LOW until CLKU goes HIGH again, duplicating the count up clock. Likewise, the Borrow output will go LOW when the circuit is in the zero state and the CLKD goes LOW.

The terminal count outputs can be used as the clock input signals to the next higher order circuit in a multistage counter, since they duplicate the clock waveforms. Multistage counters will not be fully synchronous, since there is a slight delay time difference added for each stage that is added.

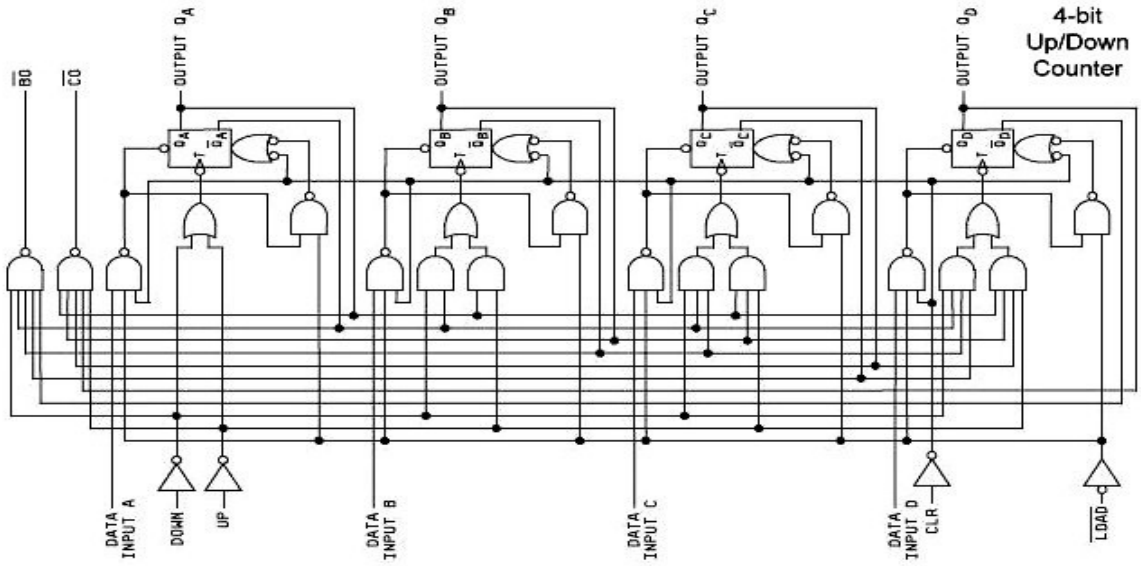
The counter may be preset by the asynchronous parallel load capability of the circuit. Information present on the parallel data inputs (P0 to P3) is loaded into the counter and appears on the outputs (Q0 to Q3) regardless of the conditions of the clock inputs when the parallel load input is LOW.

A HIGH level on the master reset input will disable the parallel load gates, override both clock inputs and set all outputs (Q0 to Q3) LOW. If one of the clock inputs is LOW during and after a reset or load operation, the next LOW-to-HIGH transition of that clock will be interpreted as a legitimate signal and will be counted. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of VCC.

Pin Diagram



Pin Configuration



Truth Table

Load	Clear			
X	1	**Clk–up**	**Clk–down**	**Mode**
1	0	X	X	Preset to zero
1	0	↑	1	Up – count
0	0	1	↑	Down count
1	0	X	X	Preset
		1	1	Stop count

Procedure Mod 10 Up & Down Counter: -

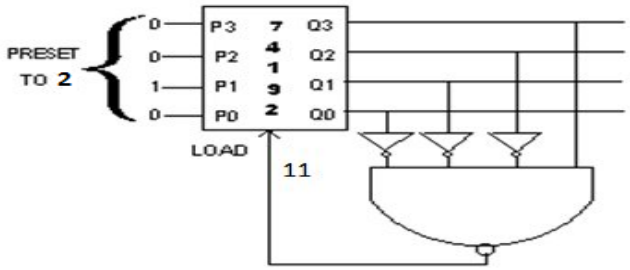
For up count clear the counter.

1. Clk-down=logic-1
2. Keep clear = logic-0 to clear the output, load can be maintained at either logic-0 or logic-1.
3. Then, first make Load=logic-1 and then Clear=logic-0, to start the counter.
4. Mono-pulses are applied at Clk-up.
5. Observe the count sequence at Q_3 Q_2 Q_1 Q_0.

For down count, clear the counter.

1. Clk-up=logic-1
2. Keep clear = 0, load = 0 to set the output.
3. Keep clear = 0 change load to 1, to start the counter.
4. Mono-pulses are applied at Clk-down.
5. Observe the count sequence at Q3 Q_2 Q_1 Q_0.

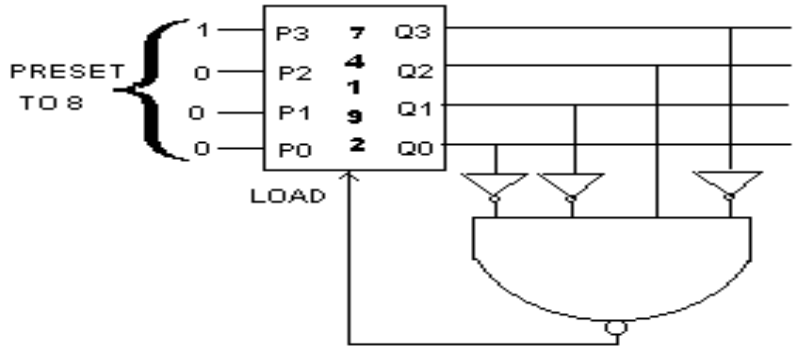
Preset table Mod 6 Up Counter
Circuit Diagram:



Procedure: -
To count from 2 to 7

1) Preset the data input to 0010.
2) Set Clk-down=logic-1
3) Set Clear=logic-0
4) Connect the load to the output of the NAND gate
5) Mono-pulses are applied at Clk-up
6) The o/p count should changes from 2 to 7, 2 to 7 & so on.

Preset table Mod 4 Down Counters:
Circuit Diagram:



Procedure for down counter

1) Preset the data input to 1000.
2) Set Clk-up=logic-1
3) Set Load=logic-0 and Clear=logic-0
4) Connect the load to the output of the NAND gate
5) Mono-pulses are applied at Clk-down
6) The o/p count should changes from 8 to 5, 8 to 5 & so on.

Assignment: Design of Multi-stage Up/Down counters using 74192 (draw ckt)

Inference:

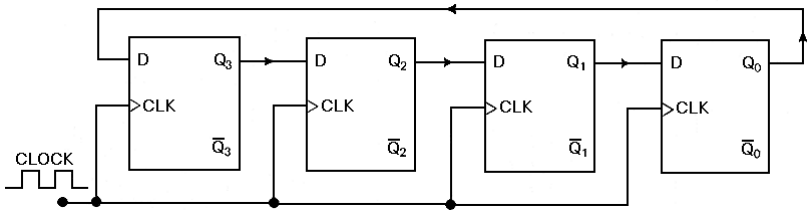
B. Design a 4bit Universal Shift Register, Verify SISO, SIPO, PISO, PIPO operations. Realize the Ring Counter and Jhonson Counter using the SR. Use the Simulator

Components Required:

Sl. No.	Components	Specification	Quantity
1	D Flip-Flops	-	4Nos
2	NOT GATE	-	1 No
3	Simulator	–	-

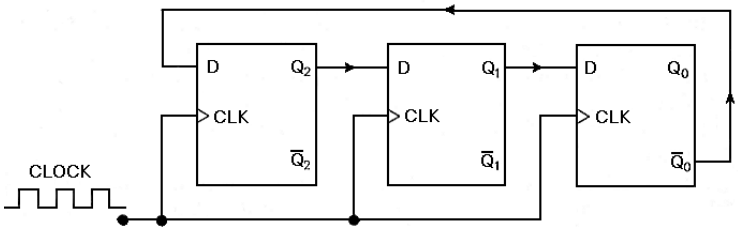
Theory : Universal Shift Register can be realized using the D Flip-Flops and basic gates. Provide following inputs: Select lines to select the mode of operation (Parallel load, shift left, shift right, hold), Serial inputs for Left shift and Right shift, Parallel load inputs, Clock to shift the data, Outputs. Perform the testing for all the modes and them build following counters.

a. Ring counter : A ring counter is a type of counter composed of a type circular shift register. The output of the last shift register is fed to the input of the first register. Ring counters are used in hardware logic design (e.g. ASIC & FPGA) to create simple finite state machines. A binary counter will require an adder circuit which is substantially more complex than a ring counter. Additionally, the worst-case propagation delay on an adder circuit will be proportional to the number of bits in the code (due to the carry propagation). The propagation delay of a ring counter will be a constant regardless of the number of bits in the code. The complex combinational logic of an adder can create timing errors which may result in erratic hardware performance. Also, ring counters with Hamming distance 2(or more) allow the detection of single bit upsets that can occur in hazardous environments.



The disadvantage of ring counters is that they are lower density codes. A binary counter can represent 2^N states, where N is the number of bits in the code, whereas an Overbank counter can represent only N states and a Johnson counter can represent only 2N states. This may be an important consideration in hardware implementations where registers are more expensive than combinational logic.

a. **JOHNSON COUNTER :** The Johnson Ring Counter or "Twisted Ring Counters", is another shift register with feedback exactly the same as the standard Ring Counter above, except that this time the inverted output Q of the last flip-flop is now connected back to the input D of the first flip-flop as shown below.



The main advantage of this type of ring counter is that it only needs half the number of flip-flops compared to the standard ring counter then it's modulo number is halved. So a "n-stage" Johnson counter will circulate a single data bit giving sequence of 2n different states and can therefore be considered as a "mod-2n counter".

Design a 4bit Universal Shift Register, Verify SISO,SIPO,PISO,PIPO operations. Realize the Ring Counter and Jhonson Counter using the SR. Use the Simulator

Truth Table: (Ring Counter)

Mode M	Clock	Inputs				Output			
		D3	D2	D1	D0	Q3	Q2	Q1	Q0
1	0	1	0	0	0				
0	1								
0	2								
0	3								
0	4								
0	5								

Truth Table: (Jhonson Counter)

Mode M	Clock	Inputs				Output			
		D3	D2	D1	D0	Q3	Q2	Q1	Q0
1	0	0	0	0	0				
0	1								
0	2								
0	3								
0	4								
0	5								
0	6								
0	7								
0	8								

Draw the Circuit Diagram of Universal Shift Register- with Mode Details

Experiment 7

Design of Mod-N Synchronous counters using JK/D/T Flip-Flops (Simulator)

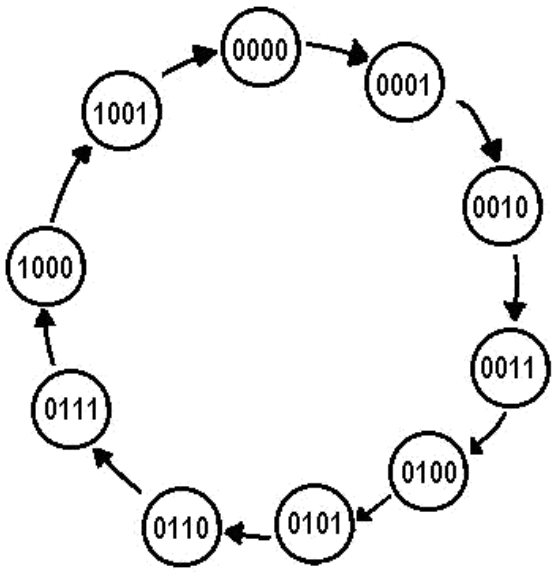
Aim: To design and test Mod-N Synchronous counters using JK/D/T Flip Flops using simulator

Components Required:

Sl. No.	Components	Specification	Quantity
1	AND GATE	-	-
2	OR GATE	-	-
3	J K/T/D Flip Flop	-	-
4	Simulator	–	-

a. Design a Decade synchronous Counter

A synchronous decade counter will count from zero to nine and repeat the sequence. The state diagram of this counter is shown in figure



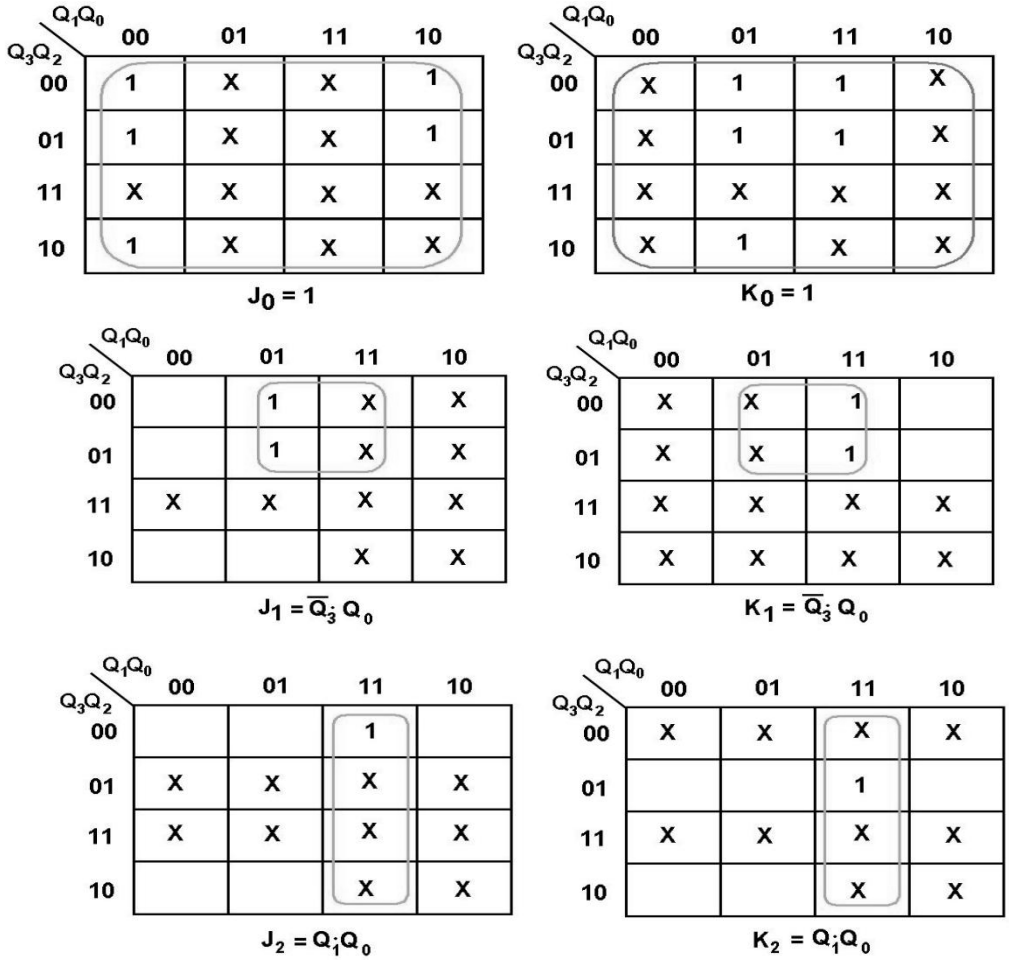
State diagram of synchronous decade counter

Since there are ten states, four JK flip-flops are required. The truth tables of present and next state for the decade counter are shown in figure. Using the excitation table of JK flip-flop and the outputs of J and K are filled.

Truth Table:

Present State				Next State				Output							
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	X	0	0	X	0	X	1	X
1	0	0	1	0	0	0	0	X	1	0	X	0	X	X	1

The Karnaugh maps of the output J_0, K_0, J_1, K_1, J_2, K_2, J_3, and K_3 are shown below:



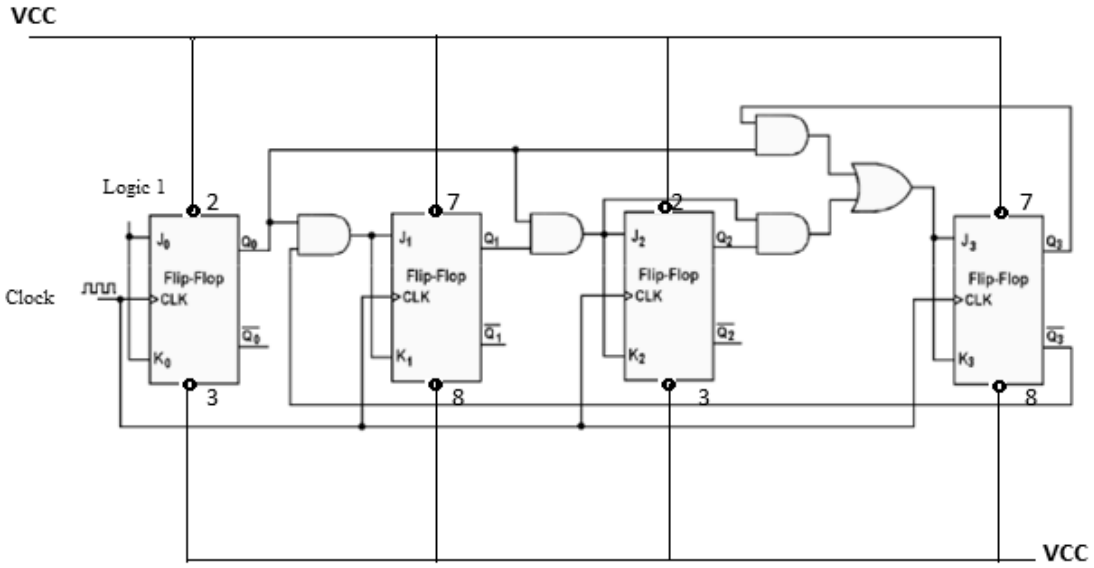
Q_3Q_2 \ Q_1Q_0	00	01	11	10
00				
01			1	
11	X	X	X	X
10	X	X	X	X

$J_3 = Q_3Q_0 + Q_2Q_1Q_0$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	X	X	X	X
10		1	X	X

$K_3 = Q_3Q_0 + Q_2Q_1Q_0$

Logic Diagram:

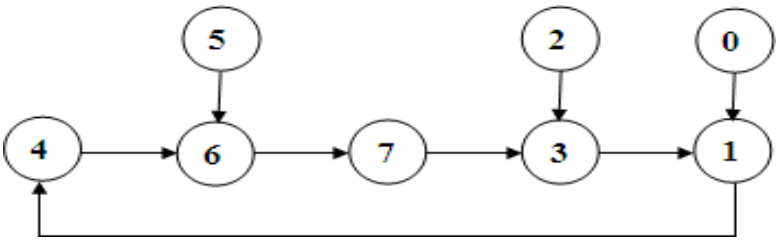


b. Design a synchronous counter for the following count sequence: 4-6-7-3-1----4-6……. (Avoid Lock out condition)

(i) Design

Step1: Writing state transition diagram.

To avoid Lock out condition, next states of unused states are sent to used state as shown in state diagram.



Here, unused states 5, 2 and 0 are forced to go into 6, 3 and 1 state respectively to avoid lockout condition.

Step 2: Writing present state, next state and transition table.

Present state			Next state			Flip-Flop inputs		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	J_2K_2	J_1K_1	J_0K_0
0	0	0	0	0	1	0X	0X	1X
0	0	1	1	0	0	1X	0X	X1
0	1	0	0	1	1	0X	X0	1X
0	1	1	0	0	1	0X	X1	X0
1	0	0	1	1	0	X0	1X	0X
1	0	1	1	1	0	X0	1X	X1
1	1	0	1	1	1	X0	X0	1X
1	1	1	0	1	1	X1	X0	X0

Step 3: Use Karnaugh maps to identify the present state logic functions for each of the inputs.

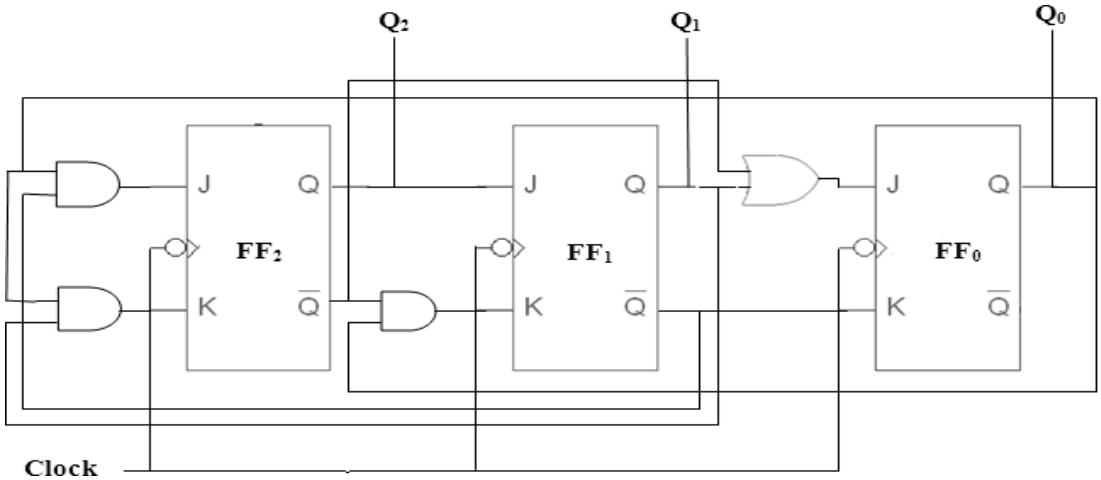
For J_2,

Q_2 \ Q_1Q_0	00	01	11	10
0	0	1	0	0
1	X	X	X	X

$J_2=Q_1'Q_0$ K2=Q1Q0 J1=Q2 K1=Q2'Q0 J0=Q2'+Q1 K0=Q1'

(ii) Implementation

Step 4: Constructing Circuit as shown in Figure



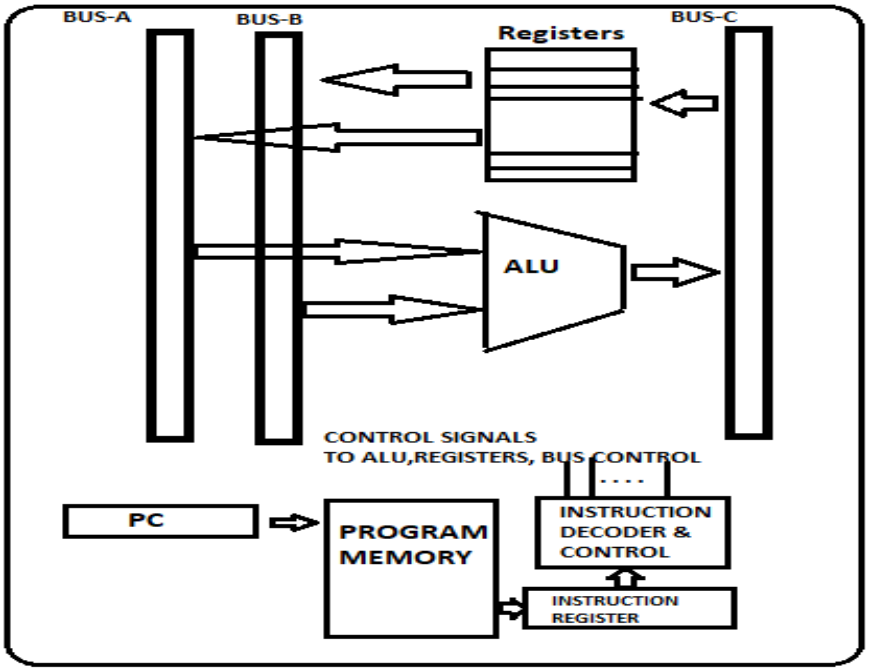
Inference:

Experiments 8,9 and 10

Design a 4-bit CPU using the LOGISIM simulator, for the following specifications.

1) Program Counter (Assume 256 locations of program/code memory)
2) Instruction Register (Assume instruction size as 16 bit)
3) General Purpose Registers (RISC type-R0-R7): Use Harvard & Multiple Bus Architecture
4) ALU (to support 4-bit integer arithmetic operations & 4-bit logical operations)
5) Memory – 1024 locations of ROM (to store instructions of size 16 bit) and 256 RAM (to store 4-bit data)
6) Implement the following instructions namely: MOV, ADD, SUB, AND, XOR, NOT.
7) Result to be displayed on 7-segment displays / reg tab of LOGISIM
8) Write multiple Assembly Language Programs and Machine Code the instructions and execute them using the designed CPU.

1.	MOV #5, R0 MOV #4, R1 MOV #A, R2 ADD R0,R1,R3 AND R3,R2,R4	2	MOV #3, R1 MOV #2, R2 ADD R1,R2,R4 ADD R1,R2,R5 OR R1,R2,R6	3	MOV #4, R1 MOV #8, R2 ADD R1,R2,R3 XOR R1,R2,R4 AND R1,R2,R6	4	MOV #a, RØ MOV #b, R1 ADD RØ,R1,R7 AND RØ,R1,R3 XOR R0,R1,R2



MULTIPLE BUS CPU ORGANIZATION

Refer the following Videos to Perform the Experiments:

Logisim 4 bit CPU Design Lecture #1
https://youtu.be/NNRLZNNECZE?si=r6Hhud-c_2lfle6_
Logisim 4 bit CPU Design Lecture #2
https://youtu.be/WQ-PJr8y3vQ?si=RYsz2rsBAyB_zOW0

Experiment 11

Aim: To Design 4bit CLA and 4bit Comparator using iterative design technique. Use Verilog HDL to build the design, simulate the design and implement on the FPGA Kit

Refer the FPGA Kit Lab Manual. (attach the datasheet)

Experiment 12

Aim: To Evaluate & Compare the different Memory and I/O standards, on the PC Motherboard.

Refer the Motherboard Manual and Motherboard and document the details (attach data sheet).

PART B (Experiential Learning)

EL
Develop any one of the following : 1. Microcontrollers based Embedded System 2. Assembly Language based software application 3. Digital System Design using Verilog HDL and Implementation on FPGA Kit

Note: The Project/Product/Program developed for EL to be presented for SEE Exam for external evaluation.

1. Title:
2. Aim:
3. Team Members with USN:
4. Components Used in Design and their specifications:

5. Block Diagram and Description (attach the print out)

- **EL Instructions:**
  1. Form a batch of 3 students from same section.
  2. Selection of Problem Statement OR assignment of Problem Statement to each batch will be done by a course course-handling faculty member under the supervision of the course coordinator & HoD.
  3. The batch of students are required to analyze the problem statement, formulate the objectives, identify the hardware & Software requirements, define deliverables, etc.
- **Evaluation Methodology:**
  1. Phase 1 after CIE1: Student team need to prepare Synopsis/ Presentation consisting of Introduction, Problem statement, formulation of the objectives, identification of the hardware & Software requirements, deliverables, and will be evaluated by course handling faculty members. Suggestions during presentation to be noted down by the team.
  2. Phase 2 after CIE 2: The progress of the problem statements is further monitored after the 2nd CIE allowing students to refine their work before the final evaluation.
  3. Phase 3 (final) before lab CIE: During the final phase, students are required to present a demonstration of the prototype, simulation results, working model/FPGA Implementation, results, etc. A consolidated report of three to four pages is to be attached to the lab manual by each student. The report may consist of a problem statement, Objectives, Hardware/Software requirements, national& International status, BOM, Results, outcomes (Paper publication, patent, product), and Skills acquired.
  4. Hardware and software required will be provided by the department.
- **SEE Evaluation:**
  1. A project exhibition will be conducted for each course. In which, each batch should present a working prototype/ Simulation model.
  2. An external examiner (industry/ academia) with a course-handling faculty member will evaluate each student on the following rubrics.

     A4 size Poster/Write up :10 Marks

     Presentation/Demonstration/Technical Depth/Innovation :30 Marks

     Viva :10 Marks

     Total : 50 Marks
  3. Additionally, demonstration of the outcome in terms of paper publication, patents, and product may be suitably rewarded.